SIJIE ZENG

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Job Objective: AI Agent Development / R&D | Class of 2027

EDUCATION

Suzhou University of Science and Technology Sep 2023 - Jun 2027

Geographic Information Science | GPA: 3.3 (Top 30% in Major) Suzhou, Jiangsu

Core Courses: Python Data Analysis, Database Principles, Spatial Data Structures, Web Programming, C Programming, Object-Oriented Programming, Scientific Computing, Big Data Analytics.

Honors: National Inspirational Scholarship, Comprehensive Third-Class Scholarship, Study Diligence Award, National Computer Design Competition (Jiangsu 1st / National 3rd Prize), National ESRI Development & Design Competition 3rd Prize, Mapmost Map Design Competition Grand Prize.

PROFESSIONAL SKILLS

- **Programming & Engineering:** Python (asyncio, data structures); FastAPI/Flask with RESTful API experience; Git, Docker, Compose orchestration; Linux; load balancing; basic Vue.
- **Databases & Middleware:** MySQL, PostgreSQL (design, SQL optimization); Redis caching; Elasticsearch; Kafka.
- **Agent & AI:** ReAct / Plan-Execute-Reflect, Tool Calling routing; Agent Harness engineering practice (tool registration & scheduling, permission gating, sandboxed execution, state management); MCP, Skills; LangChain/LangGraph, Coze, Hermes.
- **RAG & Retrieval Optimization:** vector databases & Embeddings; hybrid retrieval (BM25 + HNSW), RRF fusion, Rerank; Query Rewrite.
- **Prompt Engineering & Evaluation:** structured prompting; LLM-as-a-Judge metrics and golden-case regression; AI coding tools (Hermes, Codex, Claude Code).

INTERNSHIP EXPERIENCE

MiQro Era Technology, Shanghai Jun 2026 - Sep 2026

AI Agent Development Intern (MiqroForge / SOP Skill / MiQroGate) Shanghai

- **Internship Overview:** three parallel tracks — MiqroForge; SOP Skill digital assets (materials & quantum chemistry); MiQroGate key & usage governance (independently delivered, MIT open source).

MiqroForge — Personal AI Agent Framework (Python asyncio backend + Electron desktop)

GitHub: github.com/14790897/MiqroForge-Desktop

- **Human-in-the-loop confirmation & authorization:** per-call intent confirmation (once / remember / reject); fail-closed on timeout/error; concurrent prompts serialized; install authorization shares one entry.
- **Multi-provider search fallback chain:** classified by error type — rate limits, network and empty results keep falling back; auth failures alert and still fall back; no-key fallback (region-parallel, per-region isolated); exhausted chains surface the last provider's real failure reason; fixed the fast-mode chain-bypass defect.
- **Sandboxed execution governance (bwrap sandbox):** true-stop judged by process returncode, not task state (fixing false stops); timeouts terminate by process group; timeout configs unified into one policy.
- **Trusted delivery boundary for MCP artifacts:** SHA-256-verified atomic persistence; model context receives only path / size / hash; chunked downloads support cross-call checkpoint resume and crash-residue cleanup.
- **Document parsing & concurrency stability:** unified parsing entry point (20 format families / 41 extensions; automatic local OCR with no system dependencies); fixed lost session context and session-lock deadlocks.

SOP Skill Digital Assets (Materials Computing & Quantum Chemistry)

- **Independently delivered 2 standard job packages:** expert computation workflows codified with validation and delivery specifications; built-in cross-host environment consistency prevents version drift.
- **Represented the company at the Xinyan Zhicai AI4S Skills Hackathon (AI for Materials track):** graphene-membrane-skill ran a 5M-step NEMD simulation on the organizer's HPC, verifying complete He/CH4 sieving ($\alpha \to \infty$) and SF6 enrichment of a 12-component gas mixture.

MiQroGate — Virtual Key & Usage Governance System (MIT open source)

GitHub: github.com/sijie-Z/miqro-gate

- **Virtual key system:** full mqk_live_ lifecycle (issue → auth → route → inject → rotate/revoke); HMAC-SHA-256 storage; constant-time check; uniform 404 anti-enumeration; single key, tag routing.
- **Protocol normalization & byte-level pass-through:** 5 protocol families plus a compile-time adapter registry absorb 8 vendors / 23 products; the hot path does path normalization / credential injection / usage parsing only — everything else passes through byte-for-byte (raw-body and SSE byte-equality assertions).

- **Usage semantic normalization:** vendor token counting aligned — cached-inclusive counts always deducted; Anthropic's semantics naturally disjoint; one token maps to one billing bucket; streaming usage merged by last-frame overwrite, avoiding double-counted frames.
- **MCP gateway proxy:** verbatim JSON-RPC forwarding; circuit-broken by tool/method; two-tier service/tool ACL; no MCP SDK dependency; 40 contract tests; audit: metadata-only rows (no prompts or bodies stored).
- **Data plane & engineering:** versioned read-only routing snapshots via LISTEN/NOTIFY (lock-free reads, last-good retained); two-tier cache (L1 in-memory + L2 PostgreSQL, off by default, double opt-in) with single-flight coalescing; AES-256-GCM credentials; Ed25519-signed catalog; ArchUnit non-blocking rules.

OPEN SOURCE CONTRIBUTIONS

- **Contributed 13 merged PRs to 10 popular open-source projects:** svelte (×2), clash-verge-rev (×3), Deno, axios, Prisma, Ant Design, Storybook, pnpm, rclone and the official Vue Chinese docs — spanning JS/TS frameworks, Rust runtimes and infrastructure tooling, including a Deno use-after-close fix.
- **Featured case:** root-caused a Redis BITOP short-value regression — AVX2 SIMD dispatch missing a minimum-length gate; fix restored small-input throughput +7.1% (CE benchmark; github.com/redis/redis/pull/15323).
- Additional fixes under review for the TiDB query planner, CockroachDB, Traefik, Tauri and Effect-TS; including a KubeVela path-traversal security fix.

PROJECT EXPERIENCE

DocMind — Intelligent Document Q&A and Analysis Platform Aug 2025 - Sep 2026

Tech Stack: Python / FastAPI / Async / Elasticsearch / Redis / Kafka / MinIO / Vue 3 | GitHub: github.com/sijie-Z/DocMind-RAG

- **Project Overview:** enterprise document backend combining RAG Q&A and AI-agent analysis — upload & ingestion, hybrid retrieval, multi-turn dialogue, and multi-step tool-calling agents.
- **RAG Retrieval Design:** a dual-path retrieval engine on Elasticsearch (BM25 keyword + high-dimensional vector); RRF fuses recall results and a reranker performs a second-pass refinement to improve Top-K relevance; query rewrite converts colloquial questions into document-style queries.
- **Agent Planning & Execution Framework:** Plan-Execute-Reflect replaces single-step ReAct: the Planner builds a multi-step dependency graph; the Executor runs it topologically via per-step Function Calling; the Reflector retries locally, not fully.
- **Function Calling Tool Routing:** 25+ tools registered by schema and auto-converted to OpenAI-compatible Function Calling; tool_calls routed via registry, results fed back under the tool role; steps reuse prior outputs.
- **Context Engineering:** working / short-term / long-term memory managed in three tiers; when context exceeds the Token limit, early non-critical turns are dropped by priority or compressed into structured summaries.
- **Async High-Concurrency Processing:** a Kafka + MinIO pipeline decouples document parsing from Embedding vectorization, staying responsive under high-frequency multi-user uploads.
- **Observability:** Prometheus metrics + Langfuse tracing, 12 SSE event types, RunReport summaries.

Vector Data Distribution and Provenance System Jul 2024 - Aug 2025

Tech Stack: Python / Flask / MySQL / PostgreSQL / Redis / Docker | GitHub: github.com/sijie-Z/GeoData-Security-System

- **Project Overview:** ownership tracking and provenance for multi-stage vector & raster data — requests and approvals, QR-code watermark (LSB / DWT / histogram-shift), provenance verification.
- **Business API Development:** 100+ RESTful APIs on Flask-RESTful — authentication, requests, three-level approval flow (initial→secondary→additional review), recall proposals; approval state machine and logs.
- **Dual-Database Storage:** MySQL for business data + PostgreSQL/PostGIS for spatial data, multiple data sources managed via SQLAlchemy; 27 related tables covering user-permission many-to-many and request-approval-download joins.
- **Caching & Access Control:** Redis caching + invalidation and sliding-window rate limiting lifted API speed by 30%+; JWT Token auth with three-tier permissions, WebSocket JWT enforcement fixed identity spoofing.
- **Engineering & Deployment:** Docker Compose orchestration of 9 services (health checks, data persistence); pytest suite; GitHub Actions CI/CD pipelines.

SUMMARY

Hands-on in AI agents and LLM apps, with end-to-end RAG and agent practice; led MiqroForge's interaction loop and sandbox governance (internship), delivering an AI credential virtualization & usage gateway (MIT open source). Solid backend foundations; effective with AI coding tools (Hermes, Codex, Claude Code).